Beyond the stimulus
Individual aversive emotional reactivity in HABS

A reproducible multimodal study with first-session calibration and reserved later-session evaluation

HABS Neuro-AI Hackathon · Research report · 9 November 2026

Abstract

We tested whether physiology predicts individual aversive activation beyond stimulus context and prior-session calibration. The target S=max(0,(5−V)/4)(A−1)/8 combines post-video valence and arousal; it is an experimental proxy rather than a validated stress measure. A release audit aligned 22,974 trials from 314 identifiers. Population models and settings were learned on development identifiers; reserved participants contributed only their first session for the specified calibration. On 997 novel-to-person second-session trials from 22 reserved identifiers, the reference M1 had participant-macro MAE 0.1124, versus 0.1128 for multimodal M4. The paired improvement was -0.0004 [95% CI -0.0029, +0.0018]. The reserved comparison did not establish a clear incremental benefit. The interval excluded a gain of 0.01 on the 0–1 score for this model family. Sensor-availability, prior-physiology, nonlinear, dimensional-target and profile controls constrain interpretation.

Introduction

Predicting a label associated with an emotional video is different from predicting what a particular viewer reports. Gerster et al. showed that stimulus identity and temporal dependence can dominate apparent EEG emotion classification on FACED [1]. Prior multimodal stress classification [2] therefore does not establish that physiology adds information beyond a strong stimulus and personal-history reference. We operationalized this incremental question using the HABS release and tested it on later sessions with videos not previously shown to the evaluated person.

The scientific object is reported aversive activation. Aversive-video paradigms can produce independently measured stress responses [3], but HABS supplies no independent trial-level clinical stress or onset criterion. Physiological responses also vary within emotion categories [4]. Neither a positive prediction result nor a discrete cluster would by itself validate a stress instrument or biological phenotype.

Methods

Release and trial identity. We reconciled participant, normalized video filename and start marker against session metadata, subtracting each segment’s canonical time origin and verifying BDF onset agreement within 25 ms. The accounting was 22,974 retained, 4,213 unmatched, 2,293 missing/out-of-range ratings and 34 outside/short windows. There were 2,690 label-day corrections. Official masked HABS test records were not evaluable. Cohort 2 was primary; cohorts 1 and 4 were analysed separately because biological identity across cohort IDs is unknown.

Target and predictors. V and A are reported on 1–9 scales. The product S is zero for nonnegative valence or minimum activation. Current ratings, emotion self-reports, refined final labels and screening flags were excluded from prospective predictors. We used six regional relative EEG band powers from F3/F4/Fz and P3/P4/O1/O2, median heart rate and its within-window change, and calibrated EDA level, slope and dispersion. Welch used two-second windows with half overlap and 1–40 Hz normalization. Conservative EEG screening required two usable channels per region. ECG used a causal bandpass and plausible RR intervals; no frequency HRV was inferred from ten seconds. The supplied GSR voltage calibration was applied without endpoint clamping; invalid peripheral windows remained missing.

Quality and availability. Before final eligibility, 7,795/9,726 cohort-2 trials passed EEG screening, 3,672 had all three modalities and 2,742 EDA windows were outside calibration. The primary current-trial signals were CURATED to avoid released acausal preprocessing. Already observed J1 calibration could use the audited fallback. Models shared each comparison’s exact future grid and historical observations. Imputation, scaling and encoding were population-training-only.

Validation and models. A deterministic identifier hash, seeded 20261108, reserved one third of 76 cohort-2 identifiers with at least twenty aligned rated trials per session. The 26 reserved identifiers did not enter population fitting or selection. Five participant-grouped development folds excluded validation identifiers entirely from population fitting, allowed their separate J1 only for calibration, and scored their novel J2. Prior exposures were excluded using all recorded events, including unrated events. M0 used video, expected category/source, video-by-expected interaction, session and linear/quadratic order. M1 added a J1 median residual offset shrunk by n/(n+λ); the disabled limit and lambda in

{0,10,40} were considered. M2 added EEG, M3 ECG/EDA and M4 all modalities. Ridge strengths were {0.1,1,10,100,1000}. Physiological values and J1-centered differences were accompanied by current/past missingness and history-coverage controls. All comparisons shared the selected calibration strategy. Development selected a personal offset with λ=0.

Inference and prespecified controls. The primary endpoint was the mean of the identifiers' MAEs, with gain MAE(M1)−MAE(M4). A paired identifier bootstrap with 10,000 resamples gave percentile 95% intervals, conditional on the observed stimulus catalogue and fitted models. The engineering relevance threshold was 0.01, not a clinical threshold. Settings, source/data fingerprints and the comparison grid were frozen before reserved scoring. Controls included sensor missingness alone, J1 physiology alone, no EDA, complete-observation subgroups, physiological splines and joint circular shifts within identifier/session. Valence/arousal and an alternative composite were separate targets. A negative-valence analysis conditioned on observed arousal was explanatory, not deployable. Five-second models used only five-second quality eligibility and their own matched reference/grid. Secondary intervals were exploratory and not multiplicity-adjusted.

Results

The final primary population contained 72 development identifiers and 5,099 training trials; 48 identifiers supported grouped development validation. Reserved evaluation used 22 identifiers and 997 trials. 0 target trials had a video absent from population fitting. Four of the 26 originally reserved identifiers did not meet the final source, signal or calibration-support criteria and were not replaced. This is an internal released-identifier hold-out, not performance on the official masked challenge set. Figure 1 shows the score concentration and cohort/session structure.

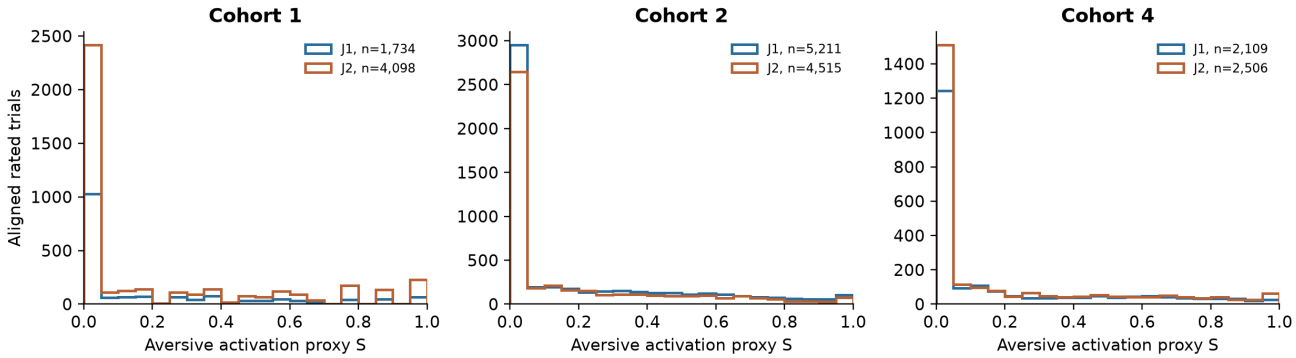


Figure 1. Proxy distributions for aligned rated observations before signal screening. Counts are descriptive trials, not independent statistical units. J1/J2 refer to reconciled recorded sessions.

Model or control	Participant-macro MAE
Stimulus/context M0	0.1157
Selected calibration M1	0.1124
EEG M2	0.1119
ECG + EDA M3	0.1131
Multimodal M4	0.1128
Availability only	0.1123
J1 physiology only	0.1127
EEG + ECG	0.1126
Physiological splines	0.1126
Cluster history M5	0.1128
Continuous PCA history	0.1128
Constant zero	0.1747

The prespecified M1−M4 gain was -0.0004 [95% CI -0.0029, +0.0018]; 45.5% of identifiers had lower M4 error. Against the availability-only control, the M4 gain was -0.0006 [95% CI -0.0029, +0.0013]; against J1 physiology alone, -0.0001 [95% CI -0.0025, +0.0019]. The spline-versus-linear increment was +0.0002 [95% CI -0.0007, +0.0011]. Figure 2 displays paired errors; Figure 7 shows the full ablation.

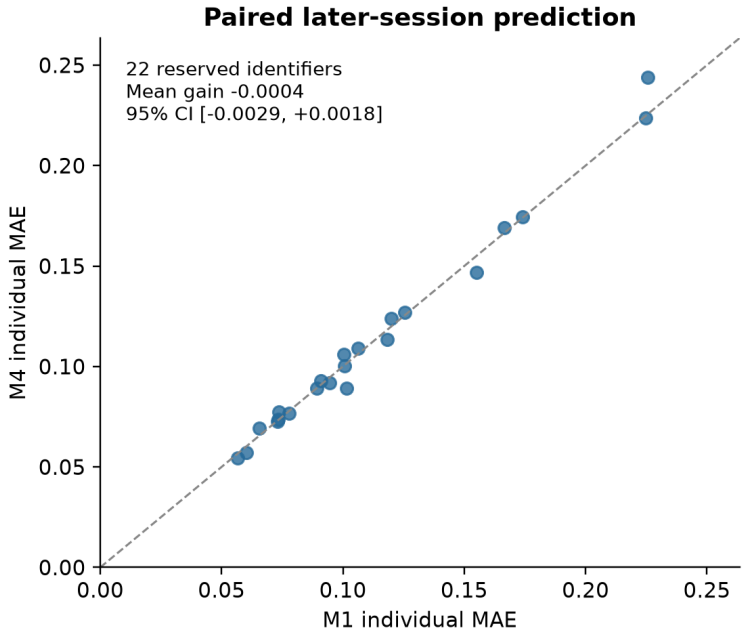


Figure 2. Each anonymous point is one reserved identifier on identical future trials. Points below the identity line favour M4. The interval resamples paired identifiers, preserving their full trial dependence.

Specificity and early windows. For negative valence conditional on observed current arousal, the physiological increment was -0.0004 [95% CI -0.0027, +0.0017]. The high-arousal descriptive matching analysis formed 123 within-identifier pairs at a fixed 0.5-point caliper. The independent five-second comparison used 23 identifiers/1,060 trials and gave +0.0005 [95% CI -0.0010, +0.0020]. Its absolute errors are not a paired five-versus-ten-second comparison, since eligibility and fitted models differ. Supporting figures show marker-level conditional differences and dose curves.

Profiles. Development selected 2 clusters without response labels. The estimated cluster-history increment M4−M5 was +0.000001 [95% CI -0.000016, +0.000018]; its increment over the dimension-matched continuous PCA-history model was -0.000011 [95% CI -0.000080, +0.000052]. The complete-case, stimulus-centered projection is descriptive and cannot establish natural biological categories.

J1/J2 complete-case descriptors were available for 10 reserved identifiers. Matched Hellinger similarity was 0.889, versus 0.775 between identifiers; the difference was +0.113 [95% CI +0.014, +0.210]. The matched value lay within the count-stratified identity-shuffle reference [0.712, 0.893]. The positive matched-versus-all-between contrast is therefore not robust against every composition/precision control. The shuffle is observational, not an exact randomized test; only ten complete-case identifiers support this description.

Separate cohorts. With cohort-2 settings fixed, the exploratory increments were cohort 1, 8 identifiers/287 trials, -0.0117 [95% CI -0.0329, +0.0050]; cohort 4, 14 identifiers/578 trials, -0.0023 [95% CI -0.0065, +0.0028]. Unknown biological overlap prevents treating these as independently sampled people or claiming cross-cohort transfer.

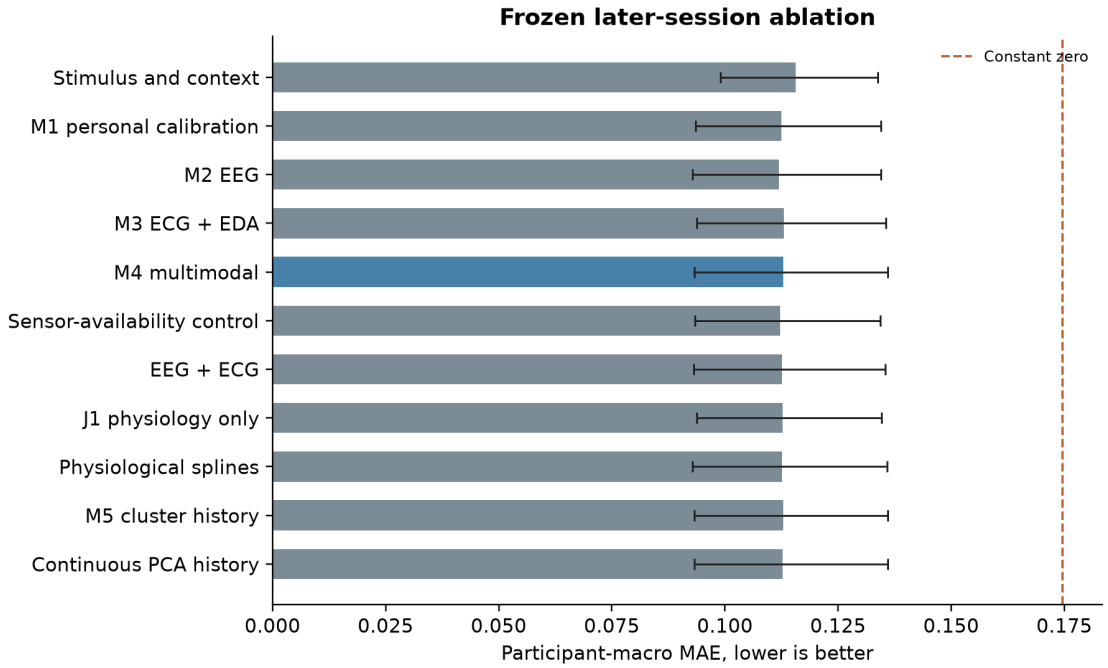


Figure 7. Frozen later-session model ablation. Error bars bootstrap identifiers. They describe each model's error; the primary difference uses a paired bootstrap. The constant-zero line is an important reference for a zero-concentrated target.

Discussion

The reserved comparison did not establish a clear incremental benefit. The interval excluded a gain of 0.01 on the 0–1 score for this model family. The inference is incremental prediction relative to explicit finite-capacity references, not conditional independence from every possible stimulus/history representation. The availability and history-only controls distinguish interpretable current-trial values from stable physiology or recording patterns. Even stable profiles can reflect baseline physiology, sensor placement or residual artifacts; improved prediction would not establish distinct stress phenotypes. Conversely, absent gain from these regularized models and simple features does not demonstrate an absence of emotional information in the raw recordings.

Several limits are substantive. The target has a mathematical zero boundary and no independent stress criterion. Subjective ratings are observations of reports, not direct access to experience. The release may already be screened, and signal-quality selection conditions the target population. Complete-case profile support is smaller than the EEG-usable sample. Simple spectral features and heuristic ECG peaks can retain artifacts. The primary interval does not include training-set, unseen-stimulus or cross-cohort identity uncertainty. Clinical validity, precisely dated stress onset and EEG-to-respiration forecasting require additional protocols. The defensible contribution is a reproducible, stimulus-aware, multisession test with explicit negative and inconclusive outcomes [5,6].

Reproducibility and data handling

Code and protocol: github.com/mathiaschebbah/habs. The source/feature fingerprints, full selected settings, development scores, participant-macro metrics, skore reports and read-only audit accompany the study. Primary frozen SHA-256: 0839097862a4fa70675d… . Raw recordings were read-only; individual annotations and report datasets remain in team-controlled storage. Published outputs contain aggregate statistics and figures. No new participant collection or clinical assessment was conducted.

References

[1] Gerster M et al. Stimulus identity rather than emotion drives EEG classification on the FACED dataset. bioRxiv, 2026. doi:10.64898/2026.06.12.731889. Preprint.

[2] Pei D et al. Identifying neurophysiological correlates of stress. Frontiers in Medical Engineering, 2024. doi:10.3389/fmede.2024.1434753.

[3] Roy S et al. Multimodal assessment of acute stress dynamics using an aversive video paradigm. Int J Clin Health Psychol, 2025;25:100607. doi:10.1016/j.ijchp.2025.100607.

[4] Siegel EH et al. Emotion fingerprints or emotion populations? Psychological Bulletin, 2018;144:343–393. doi:10.1037/bul0000128.

[5] Ernst H et al. Assessment of the human response to acute mental stress. PLOS ONE, 2023;18:e0294069. doi:10.1371/journal.pone.0294069.

[6] Ding Y et al. Dynamic Tracking of State Anxiety via Multi-Modal Data and Machine Learning. Front Psychiatry, 2022;13:757961. doi:10.3389/fpsyt.2022.757961.